

PROJECT 2 | EXPLORATORY DATA ANALYSIS

Optimal Funding Trajectories for Follow-On Investment

A Data-Driven Seed-Stage Investment Thesis Built on Crunchbase Funding Data

Every Startup Begins With a Bet

When a startup is founded, it typically raises a small seed round from angel investors or early-stage VCs to build an initial product and prove traction. The critical moment that follows is whether the company can go on to raise a follow-on round — Series A or later.

For a venture capital firm writing seed checks, correctly identifying which startups are most likely to progress is the difference between a portfolio that compounds in value — and one full of stalled, non-performing investments.

THE FUNNEL

Seed

Small early checks from angels and seed-stage VCs



Follow-On (Series A+)

Signals the company hit real milestones — worth continued investment

This analysis: ~54,000 Crunchbase companies

PROBLEM STATEMENT & GOALS

Most Seed Bets Never Pay Off

A venture capital firm struggles to reliably select seed-stage startups that will eventually secure follow-on funding — resulting in a portfolio of stalled companies. This analysis mines historical funding data to uncover the patterns that separate startups that progress from those that stall, in order to sharpen the firm's \$10M seed-stage investment criteria.

KEY QUESTIONS

- Which industries convert seed capital into follow-on rounds most often?
- Does the size of the seed check itself predict future success?
- Do geography and funding pace matter?

BASELINE TO BEAT

13.4%

of seed-stage startups ever raise a follow-on round

Goal: build data-driven criteria that beat this baseline

Trustworthy Insights Start With Clean Data

1

Source

Crunchbase startup funding dataset: company profiles, funding rounds, and amounts raised.

2

Scope

~54,000 companies worldwide, spanning industries, countries, and funding stages.

3

Cleaning

Removed blank/junk records, fixed data types, validated date logic, verified non-startup outliers by hand.

4

Cohort

Defined the seed-stage cohort (seed/angel funding > 0) and a binary follow-on success flag.

Definition used throughout: a seed-stage company “succeeds” if it goes on to raise any Series A–H round.

Five Lenses on Follow-On Success

01

Seed Check Size

Does the size of the first check predict what comes next?

02

Industry

Which sectors convert seed money into growth?

03

Geography

Where are the strongest follow-on ecosystems?

04

Funding Pace

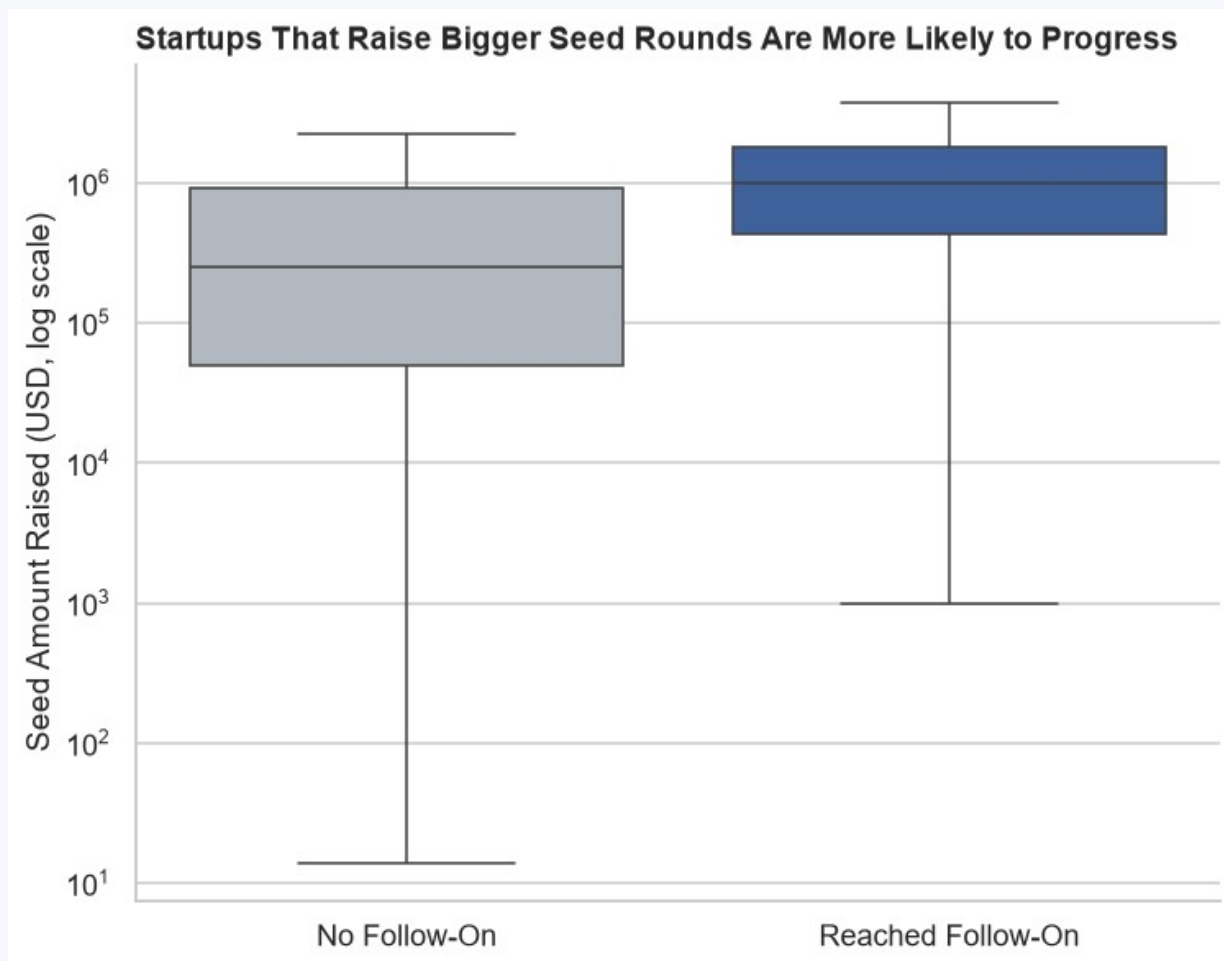
How many rounds, and how much time, does progress take?

05

Interaction Effects

How do industry and check size combine?

Bigger Seed Checks, Bigger Odds

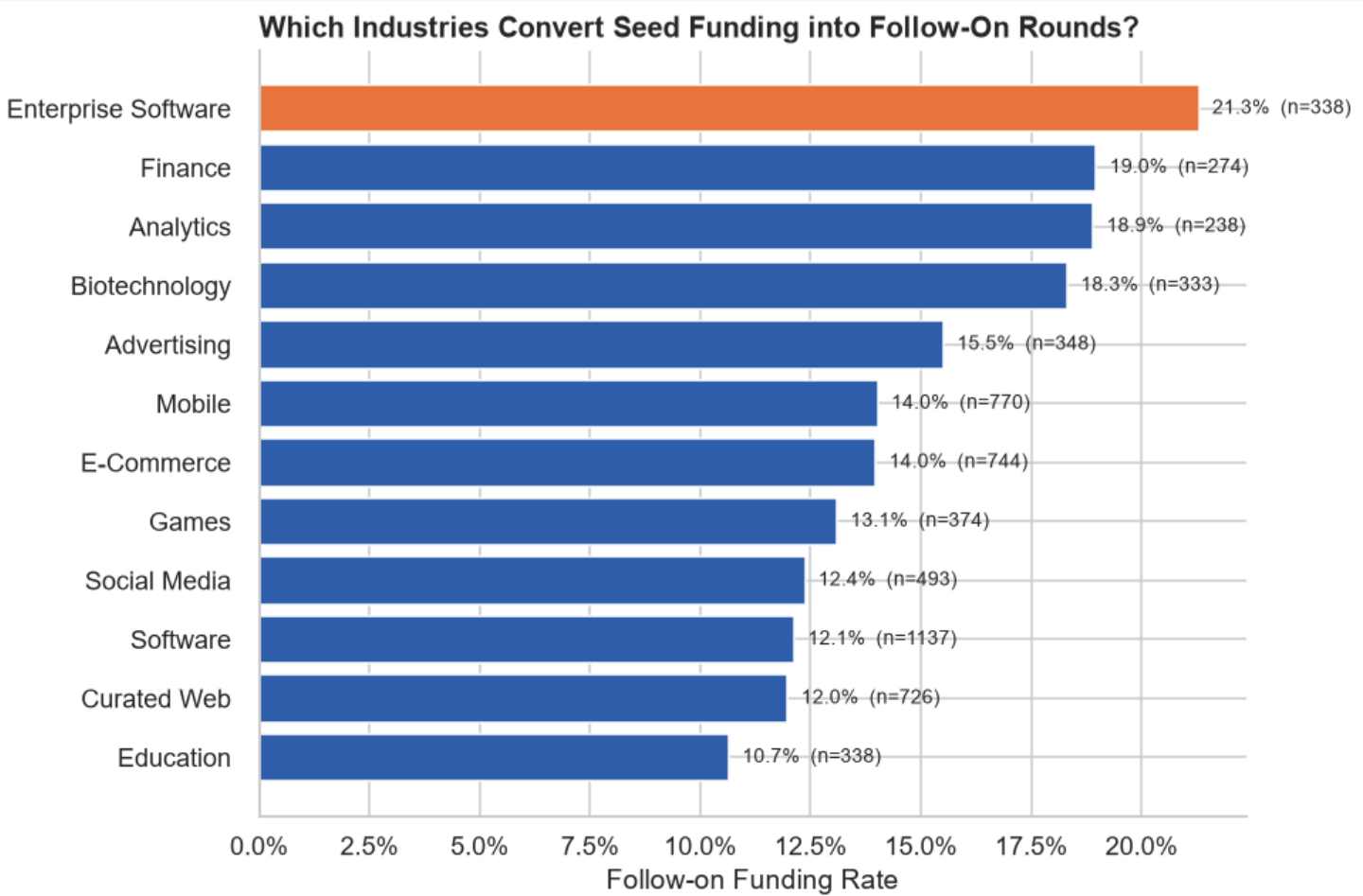


\$742K vs \$150K

Median seed round for startups that reached follow-on funding, vs. those that didn't

Startups that eventually raised a follow-on round had a seed check nearly 5x larger than those that didn't — a highly significant difference ($p < 0.001$, Mann-Whitney U test). Seed check size is the single strongest signal found in this analysis.

Not All Industries Are Created Equal



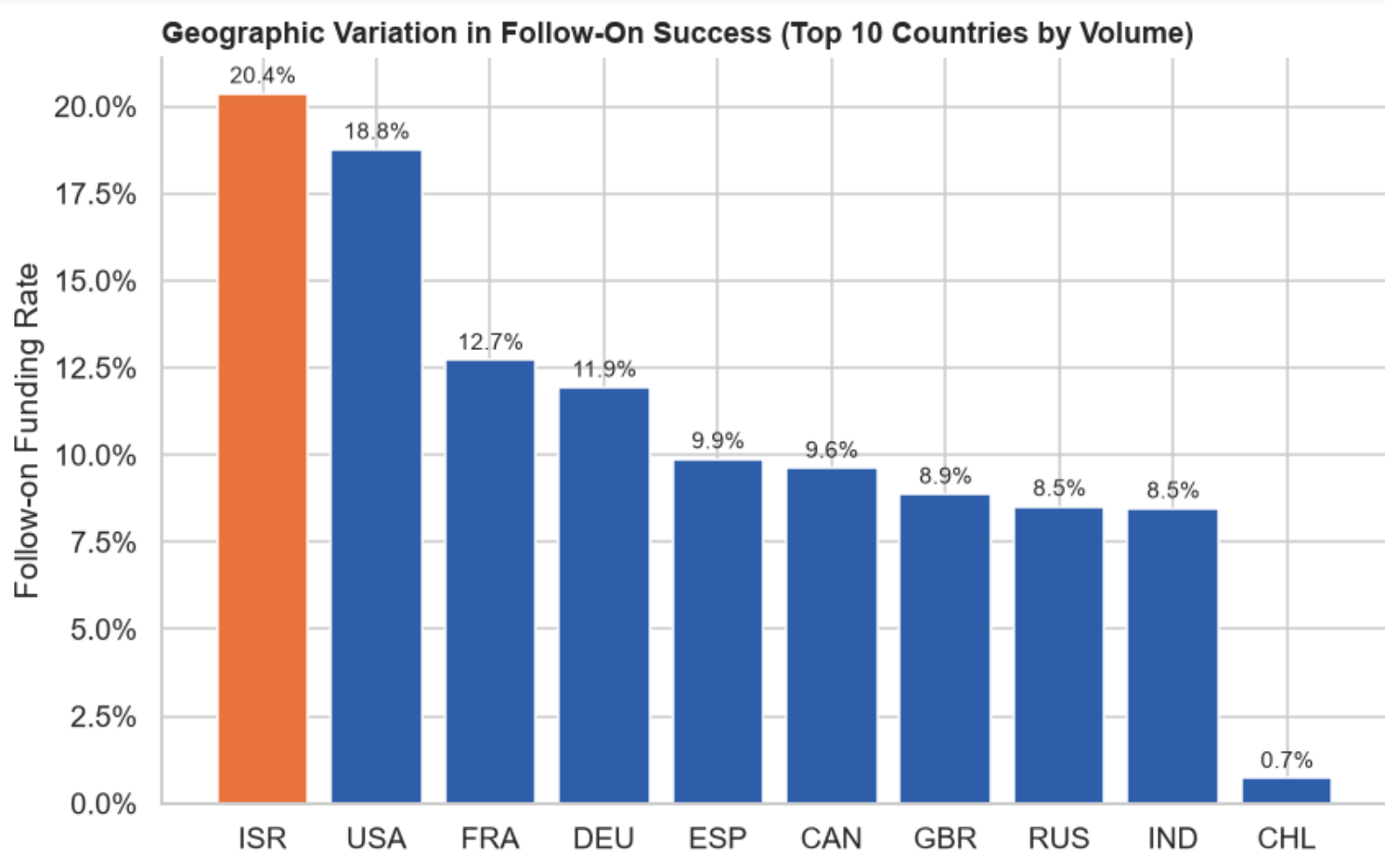
TOP CONVERTERS

Enterprise Software, Finance, Analytics & Biotechnology

convert seed funding into follow-on rounds at roughly 1.5–2x the rate of categories like Education or Curated Web — a statistically significant pattern (Chi-square, $p < 0.001$).

This directly informs where the firm's \$10M thesis should concentrate.

Location Shapes the Odds

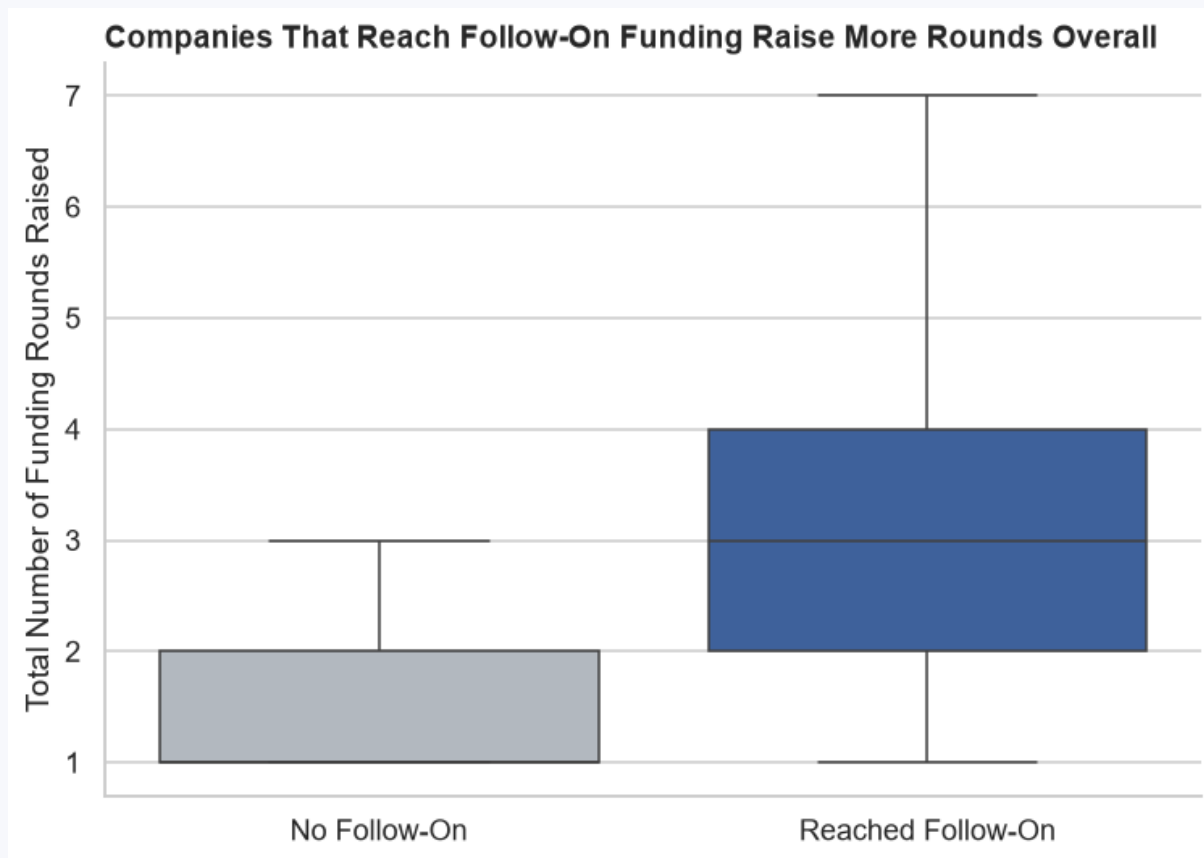


20.4%

Israel's follow-on rate — the highest of any major hub

The USA (18.8%) and Israel substantially outperform other hubs, likely reflecting deeper, more mature Series-A investor networks (Chi-square, $p < 0.001$).

Progress Takes Rounds — and Time



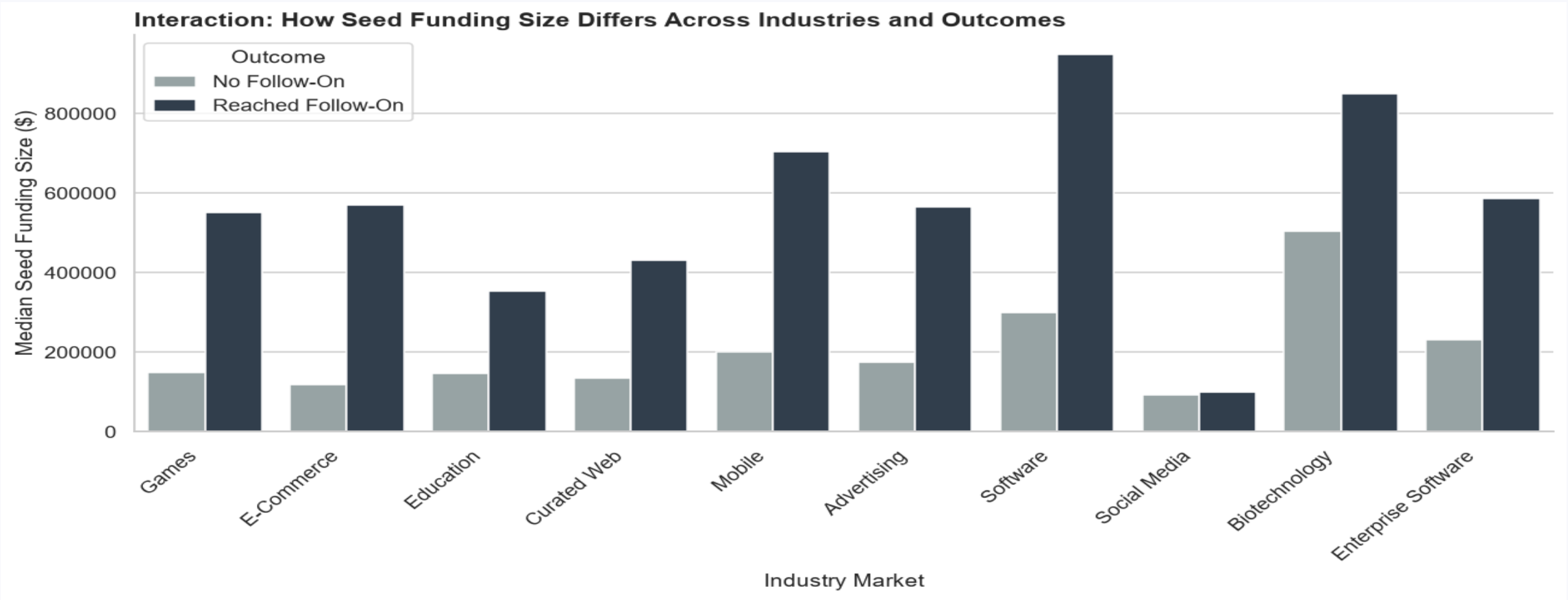
Companies that reach follow-on funding raise more rounds overall (median of 3 vs. 1), and the typical gap between a startup's seed round and its most recent round is about:

768 days

~2.1 years — the benchmark checkpoint for flagging a stalled investment

Meanwhile, how quickly a startup closes its first round after founding shows no significant relationship with follow-on success ($p = 0.23$) — a useful negative finding.

Check Size Matters Inside Every Industry



Across nearly every top industry, startups that reached follow-on funding raised a larger median seed check than those that stalled — confirming check size is a cross-sector engine, not just a one-industry effect.

What the Data Told Us

Seed check size

Strongest predictor — 5x larger for companies that progressed

Industry

Enterprise Software, Fintech, Analytics & Biotech lead by 1.5–2x

Geography

USA and Israel substantially outperform other hubs

Timeline

~768 days is the typical seed-to-latest-round window

Speed to first funding

No significant effect — safe to ignore in criteria

Turning Data Into a \$10M Thesis

- 1 Prioritize Enterprise Software, Fintech, Analytics & Biotechnology startups
- 2 Favor larger seed checks — at or above the ~\$742K historical median for success
- 3 Set an 18–24 month post-seed checkpoint to flag stalled investments early
- 4 Weight US and Israel-based teams more favorably, without excluding other regions
- 5 Do not penalize startups for a slower first raise — it isn't predictive

Next steps: this analysis is correlational and the dataset runs through ~2014 — validate against more recent funding data, and consider a causal study before finalizing investment criteria.

Thank you